Responsibility

In 2020, I joined Trevali Mining as Construction Manager at the Perkoa Zinc Mine in Burkina Faso, where I was responsible for delivering construction projects critical to maintaining uninterrupted mining operations. Reporting directly to the General Manager, I was accountable for balancing engineering, commercial, and operational priorities while coordinating multidisciplinary teams and contractors. Shortly after joining the project, I identified that the new Tailings Storage Facility (TSF II Phase III) would not be completed before the existing facility reached its remaining storage capacity. My review of survey data, construction progress, and production forecasts indicated that capacity would be exhausted by August 2020. To maintain operations, I led the development of the operational strategy and delivery of an emergency berm raise between May and September 2020 while construction of the permanent facility continued.

My role required coordinating more than 100 personnel across Engineering, Operations, Finance, Supply Chain, QA/QC, specialist consultants, and construction contractors. Although the Engineer of Record retained responsibility for engineering design and verification, I was accountable for construction planning, constructability reviews, procurement strategy, contractor management, stakeholder coordination, and successful delivery of the works.

Rather than simply reporting the risk, I took ownership of developing a practical solution. I verified survey information from both the contractor and the internal survey team, reviewed deposition rates and construction schedules, and assessed the engineering, operational, commercial, and program implications of the available options. I then developed the business case for the interim berm raise and presented my recommendation directly to the Chief Executive Officer, Chief Operating Officer, Chief Financial Officer, and Vice President. By combining engineering evidence with commercial and operational considerations, I enabled executive management to make an informed investment decision while the Engineer of Record completed and verified the engineering design.

Delivering the project during the COVID-19 pandemic required decisive leadership under rapidly changing conditions. When international travel restrictions prevented the originally selected geomembrane contractor from mobilizing to Burkina Faso, I established a multidisciplinary contractor evaluation committee comprising representatives from Engineering, Construction, Finance, and Supply Chain to identify an alternative solution. Following evaluation, I negotiated with a contractor already operating within Burkina Faso, enabling immediate mobilization and preventing significant delays. I also exercised delegated financial authority of up to US$100,000, allowing time-critical procurement decisions while remaining accountable for their engineering justification, commercial value, and compliance with company governance.

The emergency berm raise was completed for approximately US$400,000, providing approximately 257,649 m^3 of additional storage capacity and extending the operational life of the existing facility by approximately 8.9 months while TSF II Phase III progressed. This experience reinforced my understanding that professional responsibility is demonstrated not by authority alone, but by recognizing critical risks early, making balanced engineering and commercial decisions under uncertainty, earning the confidence of senior leadership, and accepting full accountability for decisions that directly affect the continuity of critical infrastructure and business operations.

Insight and Experience

Throughout my career in mining and renewable energy infrastructure, I have learned that engineering excellence is rarely achieved by applying conventional solutions. Instead, it begins by questioning whether the engineering problem has been defined correctly. This mindset has shaped my engineering judgment and was particularly demonstrated during the construction of the Ovacık Wind Power Plant in Türkiye, where I served as Site Manager between 2016 and 2019.

The project involved installing eight Nordex N131 wind turbines, requiring the transportation of 65.5-meter blades through mountainous terrain with steep gradients, narrow roads, and tight horizontal curves. The conventional engineering approach involved widening access roads, constructing retaining structures, relocating overhead utilities, and modifying the surrounding environment. While technically feasible, these measures would have increased project costs, extended the construction schedule, and created unnecessary environmental impacts.

Rather than accepting this approach, I stepped back and reconsidered the engineering constraint itself. My assessment showed that the primary challenge was not the blade length, but the turning geometry required to transport the blades safely through the most constrained sections of the route. By redefining the problem, I was able to evaluate fundamentally different engineering solutions instead of simply improving the conventional one.

I investigated the use of a hydraulic blade lifting system capable of raising the blade during transport, significantly reducing the turning radius required to negotiate difficult terrain. I completed route assessments, reviewed geometric constraints, evaluated transportation risks, and compared the technical, environmental, commercial, and operational implications of both approaches. Based on this analysis, I recommended the blade lifting solution because it minimized environmental impact, reduced unnecessary civil works, maintained the construction schedule, and achieved the greatest overall value without compromising safety. Following approval from senior management, I coordinated its implementation with transportation contractors, turbine specialists, utility providers, and local authorities.

This experience has continued to shape the way I approach engineering decisions. I learned that effective engineering is not simply about solving the problem presented but about confirming that the right problem is being solved. In many situations, the greatest opportunity for innovation comes from redefining the engineering challenge rather than developing a more complex solution.

Since then, I have applied this systems-based approach across mining and renewable energy projects. Whether evaluating construction methods, procurement strategies, or project risks, I first seek to understand the underlying engineering constraint before selecting a solution. This enables me to balance technical, commercial, environmental, and operational considerations to deliver long-term value rather than short-term technical success.

The engineering solution implemented at the Ovacık Wind Power Plant later received national media recognition for its practical approach to transporting oversized wind turbine components while minimizing environmental impact. More importantly, the experience reinforced my belief that sound engineering judgment is built through curiosity, critical thinking, and the confidence to challenge established assumptions when evidence supports a better solution.

Leadership

The greatest challenge on complex infrastructure projects is often not engineering itself, but bringing together people with different expertise, priorities, and perspectives to achieve a common objective. My experience has shown that effective leadership depends on creating clarity, maintaining open communication, and aligning diverse teams behind shared goals, particularly when projects face uncertainty.

This became especially important while serving as Construction Manager at the Perkoa Zinc Mine in Burkina Faso, where I coordinated multidisciplinary teams responsible for delivering critical mining infrastructure. Success depended on effective collaboration between the client, the Engineer of Record, contractors, survey teams, quality personnel, procurement specialists, operations staff, and specialist consultants representing different nationalities, technical disciplines, and cultural backgrounds. My role was not simply to coordinate construction activities but to create an environment where technical expertise, operational knowledge, and commercial priorities could be integrated into timely and well-informed project decisions.

One of the most demanding leadership challenges of my career arose during the COVID-19 pandemic, when international travel restrictions disrupted contractor mobilization and created significant uncertainty across the project. Rather than allowing individual disciplines to work in isolation, I established structured coordination meetings, daily progress reviews, and clear ownership of critical actions across engineering, operations, procurement, QA/QC, and construction teams. This structured leadership approach enabled engineering decisions, procurement activities, and construction operations to remain aligned despite rapidly changing site conditions. By establishing clear accountability across multiple disciplines, emerging risks were identified and resolved before they affected the construction program, allowing the emergency TSF II berm raise to be delivered without interrupting mine production. The collaborative framework established during this period continued to support effective communication and decision-making throughout the remainder of the project.

The experience also reinforced the importance of listening before making decisions. Local supervisors and site personnel frequently contributed practical knowledge of logistics, construction methods, and local operating conditions that complemented engineering documentation and technical assessments. Equally important was communicating complex technical issues in language that enabled executive management to understand both the engineering implications and the operational consequences of key decisions. Bringing together these different perspectives consistently produced stronger, more balanced outcomes than relying on technical judgement alone.

Following completion of the Burkina Faso project, I continued applying the same leadership philosophy while managing construction projects in Türkiye, including structural steel and industrial building works, before relocating to the United States. Since then, I have remained committed to continuous professional development through renewal of my PMP certification, Fellowship of the Society of Construction and Related Specialists(SCRS), membership of the Institution of Civil Engineers, and advanced training in IBM Maximo Application Suite. These experiences have reinforced my belief that effective leadership is built through continuous learning, adaptability, and collaboration across different engineering sectors and cultural environments.

Looking back, my most valuable leadership lessons have come from working in demanding environments where success depended on cooperation across multiple organizations, cultures, and technical disciplines. I believe effective leadership is not measured by authority alone, but by the ability to build trust, align diverse stakeholders, and enable sound engineering decisions that deliver safe, sustainable, and successful project outcomes.